

NAME

codata_constant_type – derived type

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **type(codata_constant_type) :: c**

C **struct codata_constant_type c**

DESCRIPTION

Defines a constant with *the name, the value, the uncertainty and the unit*.

```
type :: codata_constant_type
    character(len=64) :: name
    real(dp) :: value
    real(dp) :: uncertainty
    character(len=32) :: unit
contains
    procedure :: print
    procedure :: to_real_sp
    procedure :: to_real_dp
    generic :: to_real => to_real_sp, to_real_dp
end type codata_constant_type

struct codata_constant_type{
    char name[65];
    double value;
    double uncertainty;
    char unit[33];
};
```

EXAMPLE

```
Fortran:
    use codata
    type(codata_constant_type), pointer :: c
    c = SPEED_OF_LIGHT_IN_VACUUM
```

```
C:
    include "codata.h"
    struct codata_constant_type c;
    c = SPEED_OF_LIGHT_IN_VACUUM;
```

SEE ALSO

***codata*(3), *codata_cli*(1), *codata_version*(3), *codata_constant_type*(3)**